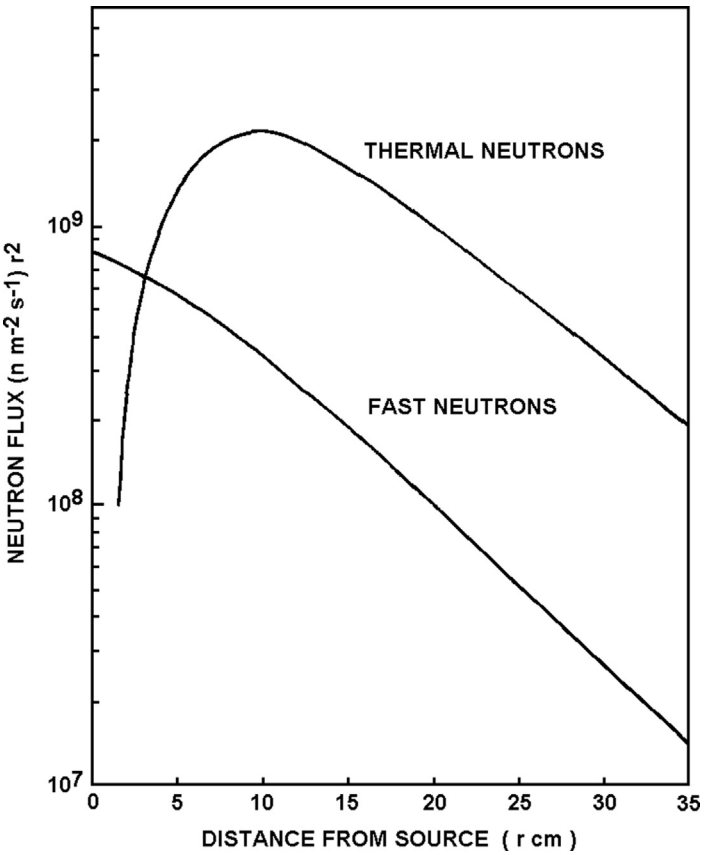


Figure 10.5 The thermalization in water of fast neutrons from a RaBe point neutron source (intensity 10^6 n s ^{1}).

Table 10.1 Distance from neutron source for maximum thermal flux

Neutron source	Neutron energy (at source)	Moderator	
		H_2O	D_2O
RaBe	Average 4 MeV	10 cm	
TD reaction	14 MeV	15 cm	
Fission	Average 2 MeV	7 cm	21 cm

10.6. INELASTIC SCATTERING

If energy is transferred not only to change the velocity of the projectile and the target but also the energy states, excitation, it is called inelastic scattering. Here a part of the kinetic energy of the projectile is transferred to the target nucleus as excitation energy without changing the values of A or Z of either target or projectile. In case the projectile is a

heavy ion it may also become excited. However, the collision of the projectile and target forming the products does result in a value of Q different than zero. For light projectiles the process can be written:

$$X_1 + X_2 \left\langle \begin{array}{ll} a & \rightarrow X_1 + X_2 + \gamma \\ & \qquad Q < 0 \\ b & \rightarrow X_1 + X_2^* \end{array} \right. \qquad (10.24a) \quad (10.24b)$$

The reaction path *a* indicates that the energy Q is emitted as a γ-ray. In reaction path *b* the Q is retained as excitation energy of the target nuclide. The latter exists in an excited state and may transform to the ground state quite rapidly or may exist for a measurable time as an isomer. An example of inelastic scattering reaction (10.24b) is the formation of an isomer of silver by the irradiation of ^{107}Ag with neutrons

$$^{107}\mathrm{Ag(n, n')}^{107m}\mathrm{Ag} \rightarrow {}^{107}\mathrm{Ag} + \gamma \qquad (10.25)$$

the half-life of ^{107m}Ag is 44 s. The energy relationships in inelastic scattering are the same as for other nuclear reactions.

10.7. DISSECTING A NUCLEAR REACTION

As an example of a nuclear transmutation reaction, let us consider the following:

$$^{14}_{7}\mathrm{N} + {}^{4}_{2}\mathrm{He} \rightarrow [{}^{18}_{9}\mathrm{F}]^* \rightarrow {}^{17}_{8}\mathrm{O} + {}^{1}_{1}\mathrm{H} \qquad (10.26)$$

For pedagogic reasons we treat this reaction as if an intermediate *compound nucleus* is formed. The compound nucleus $^{18}_{9}$F is in square brackets to indicate its transitory nature and marked with an asterisk to indicate that it is excited. Induced nuclear reactions are often written in an abbreviated manner indicating first the target and then, in parentheses, the projectile and the smaller product, followed by the major product outside the parentheses. In the case of the sample reaction, we would write ^{14}N(α,p)^{17}O The abbreviations used for ^{4}He, ^{1}H, ^{2}H (=D), ^{3}H (=T), etc., are α, p, d, t, etc. The reactions may be classified by the particles in parentheses; the sample reaction is called an (α,p) type.

The sample reaction is of historical interest since this is actually the reaction studied by Rutherford in 1919 when he produced the first induced nuclear transformation in the laboratory. A cloud chamber photograph of the reaction is shown in Figure 10.6.

From the atomic rest masses we can calculate that the change in mass for this reaction is

$$\begin{aligned} \Delta M^0 &= M_3^0 + M_4^0 - M_1^0 - M_2^0 \\ &= (16.999\,131 + 1.007\,825 - 14.003\,074 - 4.002\,603) = 0.001\,279(\mathrm{u}) \end{aligned} \qquad (10.27)$$